

SABZ

AI-ADAPTIVE TRAFFIC SIGNALS

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Green follows real demand.

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01 THE PROBLEM

Lahore added 3.7 million vehicles in a decade. Its signals never noticed.

Registered vehicles in Lahore went from about 378,000 in 2014 to 4.1 million in 2024: an elevenfold rise (Excise and Taxation Department, Punjab). The average commute is now around 34.8 minutes (Numbeo 2026 Traffic Index, via Lahore Traffic Police). Congestion costs Lahore drivers up to PKR 1,136.88 per day each (Nawaz et al., SAGE Open, 2024).

Every red light burns fuel, pumps CO2, and steals time from millions of people. The signals controlling it all run on fixed timers from another era.

A timer cannot see.

FIXED TIMER

25 seconds of green for every road, at 3 AM and at 6 PM. Empty road gets the same green as 20 waiting cars.

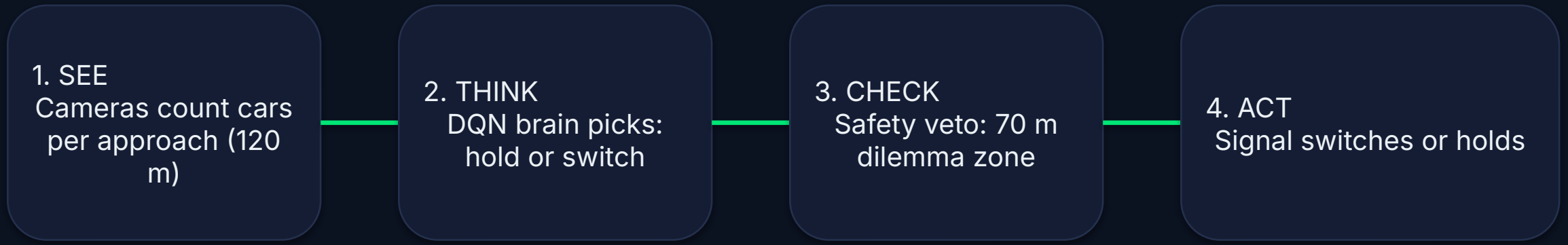
ACTUATED RULES

Simple if-then logic: if my queue is empty and someone waits, switch. It reacts. It never learns, never anticipates, never adapts to a school rush or a Friday surge.

WHAT WE NEED

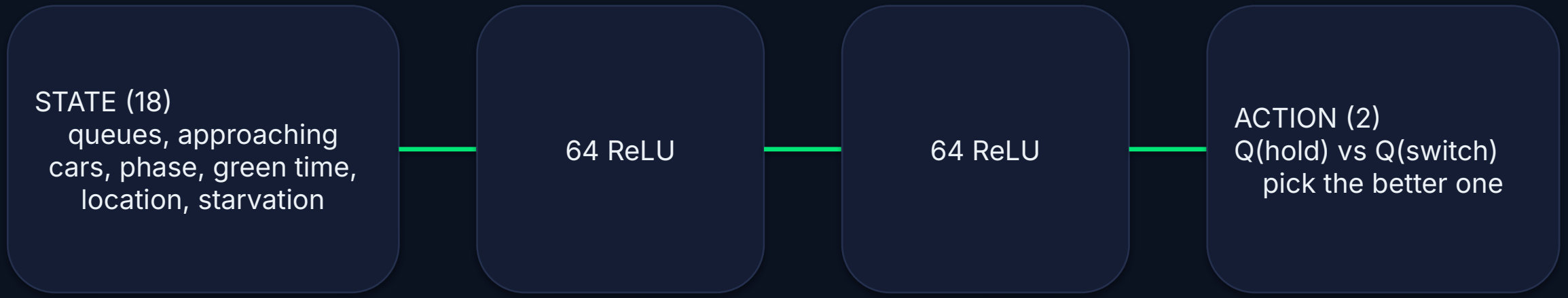
A signal that watches traffic, learns each intersection's rhythm, and gives green to the road that actually needs it. That is SABZ.

A closed loop: see, think, check, act.



The loop repeats every second. The brain is 5,506 parameters, decides in 0.02 ms on CPU, and the safety check sits outside the network where no training can touch it.

A Deep Q-Network, written from scratch.



Reward = negative total queue every second. Every car-second of waiting is felt, so the brain learns to minimize delay directly. Custom NumPy, Double DQN, prioritized replay. No PyTorch, no borrowed model.

The brain proposes. Physics disposes.

Hard rule, outside the neural network, unchangeable by training:

Every second, before any switch executes, the system scans the 70-metre dilemma zone on the green approaches. If any car is inside it, the switch is vetoed and green extends (up to 60 s max). Then 3 s yellow + 2 s all-red clearance.

With safety on, executed violations are zero by construction, and every attempted unsafe switch is logged. The AI can never learn an unsafe cutoff, because the cutoff is not its decision to make.

06 REAL DATA, HONESTLY STATED

Calibrated on real measurements. Not oversold.

NGSIM: 1,039,689 real vehicle records (3,209 vehicles, US-101). The simulator samples arrival gaps from the measured per-lane headway distribution, rescaled to each scenario's demand, so traffic has real burstiness instead of synthetic smoothness.

What it is: calibration data for arrival realism. What it is not: raw-trajectory training data, and not Lahore data. The camera pipeline is proven on real footage with YOLOv8 detection. The Lahore pilot is how real local data enters the system.

07 RESULTS

Benchmarked against the best hand-tuned controller.

3 scenarios x 3 seeds x 3,600 sim-seconds, safety on. Average delay per vehicle:

Scenario	Actuated	SABZ DQN	Result
Asymmetric	5.0 s	5.0 s	**WIN**
Balanced	11.9 s	11.9 s	**TIE (scored as loss)**
School zone	5.3 s	5.0 s	**WIN**

Best mean benchmark delay: 7.3 s. The balanced tie is disclosed, not hidden: training continues specifically to break it.

Enough slides.

Watch it think.

[SWITCH TO LIVE DEMO]

08 DEPLOYMENT VISION

One intersection in Lahore. Then every corridor.

PHASE 1: SHADOW
Brain decides, existing controller executes. Decisions logged.

PHASE 2: SUPERVISED
Live control with instant fallback to actuated/fixed.

PHASE 3: AUTONOMOUS
Full control, monitored, city can override anytime.

Cameras on existing poles. One small computer in the signal cabinet. No road digging, no GPU, no cloud dependency.

09 BUSINESS MODEL

Land one pilot. Prove it. Scale corridor by corridor.

Customer: the city traffic authority, the same body that already buys signals. Private campuses and housing societies are a faster first door.

Model: per-intersection hardware plus software license, or a managed-service fee per corridor per year. Long term: performance contracts where the city pays against measured congestion reduction.

Moat: the trained policy library per intersection type, the calibration process, and the first real deployment data from Pakistani intersections.

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Thank you.

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The people who sit in the traffic should be the people who build the fix.